

Module 5.1: AI Business Process — 30 Questions

Single Choice Questions (1-10)

Q1 According to the AI Business Development Process diagram, what is the FIRST step after business requirements?

- A. Data preprocessing
- B. Data collection
- C. Model selection and design
- D. Model training

Q2 In the requirement analysis phase, which two fundamental questions must be answered?

- A. What is the budget? and What is the timeline?
- B. What is the purpose? and What are the problems to be solved?
- C. What framework to use? and What GPU to buy?
- D. Who is the team lead? and What is the office location?

Q3 Which of the following is NOT mentioned as a recommendation system application?

- A. E-commerce product recommendation
- B. Personalized news push
- C. Video and music content recommendation
- D. Autonomous driving route recommendation

Q4 In the Computer Vision (CV) domain, which application uses datasets like KITTI, Cityscapes, and Waymo Open Dataset?

- A. Face recognition
- B. Image classification
- C. Autonomous driving
- D. Image generation

Q5 For machine translation tasks, which algorithm architecture is mentioned as the mainstream?

- A. CNN and ResNet
- B. Transformer and its variants (TransformerXL, BERT-based)
- C. YOLO and SSD
- D. PointNet and RangeNet++

Q6 When collecting data for computer vision tasks, what is the recommended approach?

- A. Use only synthetic data generated by AI
- B. Obtain diversified image/video data covering various objects, scenarios, and conditions; consider public datasets like ImageNet, COCO, or PASCAL VOC
- C. Collect only indoor images
- D. Use only black and white images for simplicity

Q7 What is the PRIMARY purpose of data cleansing in the data preparation phase?

- A. To increase the dataset size artificially
- B. To remove incorrect and unnecessary information such as duplicates, irrelevant data, and spelling errors
- C. To convert all data to binary format
- D. To encrypt the data for security

Q8 According to the module, which AI domain currently does NOT typically require feature engineering?

- A. Recommendation systems
- B. Computer Vision (CV)
- C. Natural Language Processing (NLP)
- D. Both B and C

Q9 In supervised learning, what is the purpose of the validation set?

- A. To train the model from scratch
- B. To adjust model parameters and select the best model
- C. To evaluate the final model performance only
- D. To store backup copies of the data

Q10 For an image recognition task, which model architecture family is MOST appropriate according to the module?

- A. RNN, LSTM, GRU
- B. CNN such as VGG, ResNet, and YOLO
- C. Transformer (BERT and GPT)
- D. Decision trees and random forests

Multiple Choice Questions (11-16)

Q11 Which of the following are phases in the AI Business Development Process? (Select all that apply)

- A. Business requirements
- B. Data collection
- C. Data preprocessing
- D. Model selection and design
- E. Model training
- F. Model evaluation
- G. Model deployment

Q12 Which algorithms are mentioned for recommendation systems? (Select all that apply)

- A. Collaborative filtering
- B. Content-based recommendation
- C. Knowledge-based recommendation
- D. Hybrid recommendation
- E. YouTube DNN
- F. Multimodal fusion models

Q13 Which statements about Computer Vision applications and datasets are CORRECT? (Select all that apply)

- A. Face recognition uses datasets like LFW and MS-Celeb-1M, with algorithms like CNN and FaceNet
- B. Image classification uses ImageNet, COCO, and ChestXray14 with ResNet, VGG, and Inception
- C. Autonomous driving uses KITTI, Cityscapes, and Waymo with YOLO, SSD, Faster R-CNN, PointNet, and RangeNet++
- D. Image generation is mentioned as a CV application
- E. All CV tasks require manual feature engineering

Q14 Which NLP applications and their corresponding algorithms are mentioned in the module? (Select all that apply)

- A. Chatbot: Seq2Seq, Transformer (BERT, GPT), reinforcement learning
- B. Sentiment analysis: BOW, TF-IDF, TextCNN, LSTM, bidirectional models
- C. Machine translation: Transformer, TransformerXL, BERT-based architectures
- D. Speech recognition and generation: basis for voice assistants and automatic speech translation
- E. All NLP tasks require extensive feature engineering

Q15 Which activities are part of the data preparation phase? (Select all that apply)

- A. Data collection from various sources (public datasets, crawlers, questionnaires)
- B. Data cleansing to remove incorrect and unnecessary information
- C. Feature engineering for recommendation systems (user info, object attributes, behaviors)
- D. Data labeling for supervised learning tasks
- E. Dataset splitting into training, validation, and test sets
- F. Data analysis and evaluation for quality, diversity, and balance

Q16 Which elements are considered when designing a deep learning model? (Select all that apply)

- A. Input layer
- B. Hidden layers (convolutional, recurrent, self-attention)
- C. Output layer
- D. Auxiliary layers (pooling, normalization)
- E. Hyperparameters (learning rate, batch size, regularization, dropout rate)

True or False Questions (17-20)

Q17 The AI Business Development Process starts with model training and ends with business requirements analysis.

- A. True
- B. False

Q18 For recommendation systems, feature engineering is crucial and may include user personal information, object attributes, and user behaviors.

- A. True
- B. False

Q19 In the data preparation phase, data labeling is only needed for unsupervised learning tasks.

- A. True
- B. False

Q20 The F1-Score is the arithmetic mean of precision and recall.

- A. True
- B. False

Fill in the Blank Questions (21-24)

Q21 The AI Business Development Process consists of seven phases starting from _____ requirements, followed by _____ collection, _____ preprocessing, _____ selection and design, _____ training, _____ evaluation, and finally _____ deployment.

Q22

In requirement analysis, two fundamental questions must be answered: "_____ is the purpose?" and "What are the _____ to be solved?" In the recommendation algorithm field, applications include _____ product recommendation, _____ news push, and _____ and music content recommendation. Algorithms include _____ filtering, _____-based recommendation, _____-based recommendation, and _____ recommendation.

Q23 For computer vision, applications include _____ (using datasets like LFW and MS-Celeb-1M with algorithms like CNN and FaceNet), _____ and classification (using ImageNet, COCO, and models like ResNet, VGG, Inception), _____ driving (using KITTI, Cityscapes, Waymo with YOLO, SSD, Faster R-CNN, PointNet, RangeNet++), and _____ generation. For NLP, applications include _____ (using Seq2Seq, Transformer, BERT, GPT), _____ analysis (using BOW, TF-IDF, TextCNN, LSTM), _____ translation (using Transformer, TransformerXL), and _____ recognition and generation.

Q24 In data preparation, the process includes: _____ collection from various sources, _____ to remove incorrect information, optional _____ engineering for recommendation systems, _____ for supervised learning, _____ splitting into training/validation/test sets, and _____ analysis to ensure quality and balance. In model training, _____ propagation calculates the loss function, and _____ propagation optimizes parameters based on the loss. In model deployment, steps include model _____, _____ (FP32 to FP16 and INT8), and model _____ as an API.

Drag and Drop / Matching Questions (25-30)

Q25 Match each development phase to its description:

Phase Description

- | | |
|---------------------|--|
| A. Data collection | 1. Adjust parameters and select the best model |
| B. Model training | 2. Gather raw data from various sources |
| C. Model evaluation | 3. Calculate loss and optimize parameters |
| D. Model deployment | 4. Convert, quantize, and encapsulate as API |

Write your answer as: A→?, B→?, C→?, D→?

Q26 Match each CV application to its typical dataset:

Application Dataset

- | | |
|-------------------------|--------------------------------|
| A. Face recognition | 1. ImageNet, COCO, ChestXray14 |
| B. Image classification | 2. KITTI, Cityscapes, Waymo |
| C. Autonomous driving | 3. LFW, MS-Celeb-1M |
| D. Object detection | 4. COCO, PASCAL VOC |

Write your answer as: A→?, B→?, C→?, D→?

Q27 Match each NLP task to its mentioned algorithm:

NLP Task Algorithm

- | | |
|------------------------|---|
| A. Chatbot | 1. BOW, TF-IDF, TextCNN, LSTM |
| B. Sentiment analysis | 2. Transformer, TransformerXL, BERT |
| C. Machine translation | 3. Seq2Seq, Transformer, BERT, GPT |
| D. Speech recognition | 4. Basis for voice assistants and translation |

Write your answer as: A→?, B→?, C→?, D→?

Q28 Match each evaluation metric to its definition:

Metric Definition

- | | |
|--------------|--|
| A. Accuracy | 1. Harmonic mean of precision and recall |
| B. Precision | 2. Ratio of correctly predicted samples to total samples |
| C. Recall | 3. Proportion of true positives among all positive predictions |
| D. F1-Score | 4. Proportion of positive samples correctly predicted |

Write your answer as: A→?, B→?, C→?, D→?

Q29 Match each data preparation step to its purpose:

Step Purpose

- | | |
|----------------------|---|
| A. Data cleansing | 1. Ensure quality, diversity, and balance |
| B. Data labeling | 2. Remove incorrect and unnecessary information |
| C. Dataset splitting | 3. Provide correct labels for supervised learning |
| D. Data analysis | 4. Create training/validation/test sets |

Write your answer as: A→?, B→?, C→?, D→?

Q30 Match each model architecture to its appropriate task:

Architecture Task

- | | |
|------------------------------------|-----------------------------|
| A. VGG, ResNet, YOLO | 1. Text processing and NLP |
| B. RNN, LSTM, GRU, Transformer | 2. Image recognition and CV |
| C. Deep neural network + attention | 3. Speech synthesis |
| D. CNN + pooling + normalization | 4. Recommendation systems |

Write your answer as: A→?, B→?, C→?, D→?

Scenario-Based Questions (31-35)

Q31 A retail company wants to build an AI system to recommend products to customers based on their purchase history and browsing behavior. They have collected 2 years of transaction data.

Which development phases should they prioritize?

- A. Focus only on model training with random data
- B. Data collection (already done) → Data preprocessing → Model selection (collaborative filtering/deep learning) → Training → Evaluation → Deployment
- C. Skip data preprocessing and directly train on raw transaction logs
- D. Deploy the model before any training

Q32 A startup wants to build a real-time pedestrian detection system for traffic monitoring. They need to process video streams at 30 FPS.

Which model architecture and datasets are MOST appropriate?

- A. Use BERT for text classification with IMDb dataset
- B. Use YOLO or Faster R-CNN with KITTI or Cityscapes datasets
- C. Use ResNet-50 for image classification only
- D. Use LSTM for sequence prediction

Q33 A team has collected 100,000 social media posts for sentiment analysis. They notice 15% are duplicates, 5% contain spelling errors, and 10% are in unrelated languages.

What should be their FIRST action?

- A. Immediately train a model on all raw data
- B. Perform data cleansing to remove duplicates, fix errors, and filter irrelevant content
- C. Split the data into train/test sets first
- D. Add more data without cleaning

Q34 A medical AI system detects diseases from X-rays. False negatives (missing a disease) are extremely dangerous, but false positives (false alarms) are manageable.

Which metric should be prioritized?

- A. Accuracy only
- B. Recall (sensitivity) — ability to find all positive cases
- C. Precision only
- D. Mean squared error

Q35 A company trained a PyTorch model for real-time inference. They need to deploy it on edge devices with limited memory and provide API access to mobile apps.

Which deployment steps are necessary?

- A. Model conversion to middleware format, quantization (FP32 to FP16/INT8), and API encapsulation
- B. Deploy the full FP32 PTH model directly without any optimization
- C. Only train the model; deployment is not needed
- D. Convert to Excel format for easy sharing

Advanced Concept Questions (36-40)

Q36 Why do CV and NLP currently NOT require traditional feature engineering, while recommendation systems still do?

- A. Because CV and NLP use raw data directly with deep learning models that automatically learn features, while recommendation systems benefit from explicit feature engineering for user/item understanding
- B. Because CV and NLP are simpler problems
- C. Because recommendation systems cannot use deep learning
- D. Because feature engineering is obsolete in all domains

Q37 In a spam email detection system, optimizing for high precision might reduce recall. What does this mean in practice?

- A. The system catches most spam but also marks many legitimate emails as spam
- B. The system is very sure when it marks something as spam (few false positives), but might miss some actual spam (more false negatives)
- C. The system is perfect with no errors
- D. The system randomly guesses

Q38 In the model training diagram, forward propagation calculates the loss function $C(W) = \frac{1}{2}(y - a_3)^2$. What does backward propagation compute?

- A. The predicted output a_3 only
- B. The gradients $(\partial C/\partial W_1, \partial C/\partial W_2)$ to update weights and minimize loss
- C. The input data x
- D. The activation functions g_1 and g_2

Q39 Why is model quantization (FP32 to FP16/INT8) important for deployment?

- A. It increases model accuracy
- B. It reduces model size and inference latency, enabling deployment on resource-constrained devices
- C. It makes the model require more memory
- D. It is only for training, not inference

Q40 Which statement BEST describes the iterative nature of the AI Business Development Process?

- A. The process is strictly linear and never returns to previous phases
- B. Insights from model evaluation may require returning to data collection or model selection phases for improvement
- C. Only the deployment phase can trigger changes
- D. The process is complete after one iteration

Q	Answer	Explanation
1	B	After business requirements comes data collection
2	B	Purpose and problems to be solved
3	D	Autonomous driving route is not mentioned as recommendation
4	C	KITTI, Cityscapes, Waymo are for autonomous driving
5	B	Transformer and variants for machine translation
6	B	Diversified data with public datasets
7	B	Remove incorrect/unnecessary information
8	D	CV and NLP do not require feature engineering currently
9	B	Validation set adjusts parameters and selects best model
10	B	CNN (VGG, ResNet, YOLO) for image recognition
11	A, B, C, D, E, F, G	All seven phases
12	A, B, C, D, E, F	All mentioned algorithms
13	A, B, C, D	E is false; CV doesn't require manual feature engineering
14	A, B, C, D	E is false; NLP doesn't require extensive feature engineering
15	A, B, C, D, E, F	All are data preparation steps
16	A, B, C, D, E	All are model design elements
17	B - False	Process starts with requirements, ends with deployment
18	A - True	Feature engineering includes user info, objects, behaviors
19	B - False	Labeling is for supervised learning, not unsupervised
20	B - False	F1 is harmonic mean, not arithmetic mean
21	business, data, data, model, model, model, model	Seven phases
22	What, problems, e-commerce, personalized, video, collaborative, content, knowledge, hybrid	Requirement analysis and recommendation
23	security/face recognition, image recognition, autonomous, image, chatbot, sentiment, machine, speech	CV and NLP applications
24	data, cleansing, feature, labeling, dataset, data, forward, backward, conversion, quantization, encapsulation	Data prep and model training/deployment
25	A→2, B→3, C→1, D→4	Collection=gather, Training=loss/optimize, Evaluation=adjust/select, Deploy=convert/quantize/API
26	A→3, B→1, C→2, D→4	Face=LFW/MS-Celeb, Classification=ImageNet/COCO, Driving=KITTI/Cityscapes, Detection=COCO/PASCAL
27	A→3, B→1, C→2, D→4	Chatbot=Seq2Seq/Transformer, Sentiment=BOW/TF-IDF, Translation=Transformer, Speech=voice assistants
28	A→2, B→3, C→4, D→1	Accuracy=correct/total, Precision=TP/all positive predictions, Recall=TP/all actual positives, F1=harmonic mean
29	A→2, B→3, C→4, D→1	Cleansing=remove bad data, Labeling=supervised labels, Splitting=train/val/test, Analysis=quality check
30	A→2, B→1, C→4, D→2	VGG/ResNet/YOLO=CV, RNN/LSTM/Transformer=NLP, DNN+attention=Recommendation, CNN+pooling=CV layers
31	B	Proper sequence with recommendation-focused model selection
32	B	YOLO/Faster R-CNN for real-time detection, KITTI/Cityscapes for training
33	B	Data cleansing is the first step before any modeling
34	B	Recall prioritizes finding all positive cases (minimizing false negatives)
35	A	Conversion, quantization, and API encapsulation are deployment steps

36	A	Deep learning auto-learns features in CV/NLP; recommendation benefits from explicit features
37	B	High precision = few false positives, but may miss some spam (lower recall)
38	B	Backprop computes gradients to update weights
39	B	Quantization reduces size and latency for edge deployment
40	B	Process is iterative; evaluation insights drive improvements in earlier phases

Module 5.2: Large Model Business Process — 50 Questions

Single Choice Questions (1-15)

Q1 According to the Large Model Business Development Process diagram, what phase comes immediately AFTER "Large model pre-training"?

- A. Data collection
- B. Large model fine-tuning and AI alignment
- C. Model evaluation
- D. Model deployment

Q2 GPT-3 has approximately how many parameters?

- A. 1.7 billion
- B. 17 billion
- C. 175 billion
- D. 1.75 trillion

Q3 Which of the following is NOT a valid LLaMA2 model size?

- A. 7B
- B. 13B
- C. 30B
- D. 70B

Q4 Why do Transformer-based large models require so many parameters?

- A. Because they use only feed-forward layers
- B. Because the self-attention mechanism allows each position to independently consider all other positions, capturing long-distance dependencies
- C. Because they eliminate all recurrent connections
- D. Because they use smaller datasets

Q5 How many training tokens did GPT-3 use?

- A. 30 billion
- B. 300 billion
- C. 3 trillion
- D. 30 trillion

Q6 The original GPT-3 data was 45TB, but after filtering only remained:

- A. 45GB
- B. 450GB
- C. 570GB
- D. 5.7TB

Q7 Which of the following is NOT mentioned as a data type for large model training?

- A. Texts
- B. Images and videos
- C. Audio
- D. Binary executable files

Q8 During pre-training, what hardware configuration is typically used?

- A. A single CPU
- B. A supercomputer with thousands of powerful GPUs and fast networks
- C. Only cloud TPUs
- D. Raspberry Pi clusters

Q9 In data parallelism, how many full model copies are maintained?

- A. One shared model across all nodes
- B. Each node keeps a full model copy
- C. No model copies; weights are computed on-the-fly
- D. Only the master node has the model

Q10 In tensor parallelism, a tensor is split into N blocks along one dimension, and each device holds:

- A. The full tensor
- B. $1/N$ of the tensor
- C. N copies of the tensor
- D. Only the gradients

Q11 In pipeline parallelism with 4 NPUs, how are model parameters divided?

- A. Randomly across all devices
- B. By layer, placed on device 0, 1, 2, and 3 in sequence
- C. All parameters on device 0 only
- D. Duplicated on every device

Q12 In Supervised Fine-Tuning (SFT), what happens to the output layer from the source model?

- A. It is kept unchanged
- B. It is removed because it is closely tied to the source dataset's labels
- C. It is duplicated for the target model
- D. It is frozen but not removed

Q13 In Adapter Tuning, what is the key advantage?

- A. The entire model is retrained
- B. Only task-specific adapter parameters are trained; the main model is frozen
- C. It requires more compute than full fine-tuning
- D. It removes all pre-trained knowledge

Q14 What is the main difference between Prefix Tuning and Prompt Tuning?

- A. Prefix tuning adds virtual tokens to all layers; Prompt tuning adds tokens only to the input layer
- B. They are identical methods with different names
- C. Prompt tuning trains all parameters; Prefix tuning freezes everything
- D. Prefix tuning is only for computer vision

Q15 In LoRA, how are matrices A and B initialized?

- A. Both are random Gaussian
- B. A is random Gaussian; B is zero matrix
- C. Both are zero matrices
- D. A is zero; B is random Gaussian

Multiple Choice Questions (16-25)

Q16 Which capabilities make large models superior to traditional models? (Select all that apply)

- A. Better learning capability due to more parameters and complex structure
- B. Better generalization capability on new data
- C. Stronger multi-task processing capability
- D. Require less training data
- E. Always smaller in size

Q17 Which data sources are mentioned for GPT-3 training? (Select all that apply)

- A. CommonCrawl
- B. WebText2
- C. Books1 and 2
- D. Wikipedia
- E. ArXiv and StackExchange

Q18 Which data sources are mentioned for LLaMA-65B? (Select all that apply)

- A. CommonCrawl
- B. C4
- C. GitHub
- D. Wikipedia
- E. ArXiv and StackExchange

Q19 Which are mentioned as sources for large model training data? (Select all that apply)

- A. Public data on the Internet (news, social media, blogs)
- B. Professional databases and institutional datasets
- C. Internal enterprise data
- D. User-generated content (comments, ratings, feedback)
- E. Only government classified data

Q20 Which types of parallelism are discussed for large model training? (Select all that apply)

- A. Data parallelism
- B. Tensor parallelism
- C. Pipeline parallelism
- D. Model parallelism (general term)
- E. Only single-GPU training

Q21 Which parameter-efficient fine-tuning methods are mentioned? (Select all that apply)

- A. Adapter Tuning
- B. Prefix Tuning
- C. Prompt Tuning
- D. LoRA (Low-Rank Adaptation)
- E. Full-parameter fine-tuning

Q22 What are the 3H principles for alignment? (Select all that apply)

- A. Helpfulness
- B. Honesty
- C. Harmlessness
- D. High-speed
- E. High-accuracy

Q23 Which are the three main steps of RLHF? (Select all that apply)

- A. Supervised fine-tuning (SFT)
- B. Reward model training
- C. Reinforcement learning with PPO
- D. Unsupervised pre-training only
- E. Data collection without human feedback

Q24 Which dimensions are used for large model evaluation? (Select all that apply)

- A. Knowledge and capability
- B. Ethics and security
- C. Vertical domains
- D. Only accuracy
- E. Only speed

Q25 Which evaluation benchmarks are mentioned? (Select all that apply)

- A. GLUE and SuperGLUE
- B. GSM8K and MMLU
- C. BLUE-bench
- D. Adversarial Robustness Benchmarks
- E. LLM Ethics Benchmarks

True or False Questions (26-32)

Q26 Large models based on Transformer use the self-attention mechanism, which brings a large number of parameters.

- A. True
- B. False

Q27 GLM-130B was trained on 400 billion tokens.

- A. True
- B. False

Q28 In data parallelism, only node 0 keeps a full model copy and all other nodes compute gradients only.

- A. True
- B. False

Q29 In tensor parallelism, row-wise weight partitioning requires splitting input data X, while column-wise weight partitioning does not require splitting X.

- A. True
- B. False

Q30 During SFT, the target model copies all design elements and parameters from the source model except the output layer, and a new output layer is randomly initialized.

- A. True
- B. False

Q31 RLHF uses human feedback to refine the model, ensuring it performs tasks more aligned with human targets, aspirations, and values.

- A. True
- B. False

Q32

BLEU is commonly used in machine translation tasks and checks how many n-grams match between translated text and reference text.

- A. True
- B. False

Fill in the Blank Questions (33-38)

Q33 The Large Model Business Development Process consists of: _____ requirements, _____ collection, _____ preprocessing, large model _____ and design, large model _____, large model _____ and AI _____, model _____, and model _____.

Q34 Large models often have _____ or even more parameters and can be hundreds of _____ or larger. GPT-3 has about _____ billion parameters. LLaMA2 comes in three sizes: _____, _____, and _____. The self-mechanism allows the model to independently consider each position in the input sequence and capture _____ dependencies. For complex, high-dimensional data, more parameters help the model fit better, reduce _____, and improve training performance.

Q35 The LLM training process consists of four stages: (1) _____ using >1 trillion tokens from internet data to build a base model, (2) _____ Fine-Tuning using 10K-100K demonstration data (prompt, response), (3) _____ Modeling using 100K-1M comparison data to rank responses, and (4) _____ using 10K-100K prompts to generate responses that maximize the reward. The final output is the _____ model.

Q36 In _____ parallelism, the training dataset is split into N parts, each on a separate computing node. Each node keeps a _____ model copy and calculates gradients using local data. All nodes send gradients to node _____, which combines them and sends the final result back. In _____ parallelism, a tensor is split into N blocks along one dimension, each device holds _____ of the tensor, requiring extra _____ to ensure correct computation. In _____ parallelism, model parameters are divided by _____ and placed on devices in sequence.

Q37 In LoRA, the pre-trained model's matrix parameters are _____ and replaced with matrices _____ and _____. Matrix _____ starts with random _____ values, and matrix _____ starts as a _____ matrix to keep the bypass matrix at zero initially. During training, only _____ and _____ are updated. At the end, the parameters of _____ and the pre-trained model are _____.

Q38 RLHF stands for _____ from _____. The reward model training involves: randomly selecting questions, generating multiple _____ using the SFT model, having labeling personnel _____ the responses, and using these _____ results to train the reward model. The RM model takes an input and gives a _____ to rate response quality. For each training pair, it adjusts parameters to ensure _____ responses get higher scores than _____ responses.

Drag and Drop / Matching Questions (39-44)

Q39 Match each model to its parameter count:

Model Parameters

- | | |
|-------------|-----------------|
| A. GPT-3 | 1. 7B, 13B, 70B |
| B. LLaMA2 | 2. 175 billion |
| C. GLM-130B | 3. 130 billion |

Write your answer as: A→?, B→?, C→?

Q40 Match each LLM training stage to its typical data size:

Stage Data Size

- | | |
|---------------------------|-----------------------|
| A. Pre-training | 1. 10K-100K |
| B. Supervised Fine-Tuning | 2. >1 trillion tokens |
| C. Reward Modeling | 3. 100K-1M |
| D. Reinforcement Learning | 4. 10K-100K |

Write your answer as: A→?, B→?, C→?, D→?

Q41 Match each method to its key mechanism:

Method Mechanism

- | | |
|-------------------|---|
| A. Adapter Tuning | 1. Add virtual tokens before input, freeze main model |
| B. Prefix Tuning | 2. Add small trainable modules to layers, freeze main model |
| C. Prompt Tuning | 3. Add trainable embeddings only to input layer |
| D. LoRA | 4. Use low-rank matrices A and B alongside frozen weights |

Write your answer as: A→?, B→?, C→?, D→?

Q42 Match each parallelism type to its description:

Parallelism Description

- | | |
|-------------------------|--|
| A. Data Parallelism | 1. Split tensor into blocks along one dimension |
| B. Tensor Parallelism | 2. Split dataset across nodes, each has full model |
| C. Pipeline Parallelism | 3. Split model layers across devices sequentially |

Write your answer as: A→?, B→?, C→?

Q43 Match each metric to its primary use case:

Metric Use Case

- | | |
|----------|--|
| A. BLEU | 1. Machine translation, n-gram matching |
| B. NIST | 2. Improved BLEU with information weighting |
| C. ROUGE | 3. Focus on recall, longest common subsequence |

Write your answer as: A→?, B→?, C→?

Q44 Match each benchmark to its evaluation focus:

Benchmark Focus

- | | |
|---------------------------|--|
| A. GLUE/SuperGLUE | 1. Natural language understanding |
| B. GSM8K | 2. Math problem-solving |
| C. MMLU | 3. 57 disciplines, comprehensive knowledge |
| D. BLUE-bench | 4. Biomedical field, medical Q&A |
| E. Adversarial Robustness | 5. Resistance to malicious inputs |

Write your answer as: A→?, B→?, C→?, D→?, E→?

[Scenario-Based Questions \(45-48\)](#)

Q45 A company wants to pre-train a 70B parameter model from scratch. They have a cluster of 1000 GPUs with high-speed interconnects.

Which training strategy combination should they use?

- A. Only data parallelism
- B. Data parallelism + tensor parallelism + pipeline parallelism
- C. Only single-GPU training
- D. CPU-only training

Q46 A startup has limited compute budget (4 GPUs) but wants to fine-tune a 13B parameter model for a customer service chatbot. They have 50K labeled conversation examples.

Which fine-tuning approach is MOST cost-effective?

- A. Full-parameter fine-tuning of all 13B parameters
- B. LoRA with rank $r=16$, training only the low-rank matrices
- C. Train from scratch without pre-trained model
- D. Use only prefix tuning with 1000 tokens

Q47 A company deployed an LLM for medical advice. Users report that the model sometimes gives harmful or misleading health information.

Which approach should they implement to improve safety?

- A. Increase the model size to 175B parameters
- B. Implement RLHF with human medical experts providing feedback on response safety and accuracy
- C. Remove all medical terms from training data
- D. Switch to a rule-based system instead of AI

Q48 A research team evaluates a large model on public benchmarks and gets excellent scores. However, in real-world enterprise deployment, the model performs poorly on domain-specific tasks and shows bias.

What does this demonstrate?

- A. Public benchmarks perfectly predict real-world performance
- B. The lack of unified industry standards and enterprise-specific evaluation; need for domain-specific benchmarks and adversarial testing
- C. The model is too small
- D. Evaluation is unnecessary

Advanced Concept Questions (49-50)

Q49 Why is reinforcement learning more effective than supervised learning for large language model alignment?

- A. Supervised learning gives feedback on individual tokens; reinforcement learning gives feedback on the whole output, better addressing hallucination and reward accumulation in multi-round dialogs
- B. Reinforcement learning is always faster to train
- C. Supervised learning cannot be used for language models
- D. Reinforcement learning eliminates the need for any data

Q50 What are the PRIMARY challenges in large model evaluation mentioned in the module?

- A. Diverse capabilities make comprehensive evaluation difficult; lack of unified industry standards; lack of enterprise application standards; rapid model updates causing heavy manual workload; difficulty ensuring no overlap between test data and training data
- B. Evaluation is too easy and requires no effort
- C. Only one benchmark is needed for all models
- D. Manual evaluation is always perfect and unbiased

Q	Answer	Explanation
1	B	After pre-training comes fine-tuning and alignment
2	C	GPT-3 $\approx$ 175 billion parameters
3	C	LLaMA2 sizes: 7B, 13B, 70B (no 30B)
4	B	Self-attention considers all positions independently
5	B	GPT-3 used 300 billion tokens
6	C	45TB $\rightarrow$ 570GB after filtering
7	D	Binary executables not mentioned
8	B	Supercomputer with thousands of GPUs
9	B	Each node keeps full model copy in data parallelism
10	B	Each device holds 1/N of tensor
11	B	By layer, on devices 0-3 sequentially
12	B	Output layer removed (tied to source labels)
13	B	Only adapter parameters trained; main model frozen
14	A	Prefix tuning = all layers; Prompt tuning = input only
15	B	A = random Gaussian; B = zero matrix
16	A, B, C	D, E false; large models need more data, are larger
17	A, B, C, D, E	All mentioned for GPT-3
18	A, B, C, D, E	All mentioned for LLaMA-65B
19	A, B, C, D	E false; not only government data
20	A, B, C	D is general term; E false
21	A, B, C, D	E is not parameter-efficient
22	A, B, C	3H = Helpfulness, Honesty, Harmlessness
23	A, B, C	D, E are not RLHF steps
24	A, B, C	D, E false; evaluation is multi-dimensional
25	A, B, C, D, E	All benchmarks mentioned
26	A - True	Self-attention brings many parameters
27	A - True	GLM-130B trained on 400B tokens
28	B - False	ALL nodes keep full model copy
29	A - True	Row-wise splits X; column-wise doesn't
30	A - True	SFT removes output layer, randomizes new one
31	A - True	RLHF aligns with human values
32	A - True	BLEU for translation, n-gram matching
33	business, data, data, selection, pre-training, fine-tuning, alignment, evaluation, deployment	Seven phases
34	billions, gigabytes, 175, 7B, 13B, 70B, attention, long-distance, bias	Large model scale and mechanism
35	Pre-training, Supervised, Reward, Reinforcement, Learning, final	Four-stage LLM training
36	data, full, 0, tensor, 1/N, communication, pipeline, layer	Three types of parallelism
37	frozen, A, B, A, Gaussian, B, zero, A, B, BA, merged	LoRA mechanism
38	Reinforcement, Learning, Human, Feedback, responses, rank, ranking, score, high-quality, low-quality	RLHF process
39	A $\rightarrow$ 2, B $\rightarrow$ 1, C $\rightarrow$ 3	GPT-3=175B, LLaMA2=7/13/70B, GLM=130B
40	A $\rightarrow$ 2, B $\rightarrow$ 1, C $\rightarrow$ 3, D $\rightarrow$ 4	Pre-training=trillion, SFT=10K-100K, RM=100K-1M, RL=10K-100K

41	A→2, B→1, C→3, D→4	Adapter=modules, Prefix=virtual tokens, Prompt=input embeddings, LoRA=low-rank matrices
42	A→2, B→1, C→3	Data=split dataset, Tensor=split tensor, Pipeline=split layers
43	A→1, B→2, C→3	BLEU=translation, NIST=improved BLEU, ROUGE=recall focus
44	A→1, B→2, C→3, D→4, E→5	All benchmarks matched correctly
45	B	Large models need all three parallelism types
46	B	LoRA is most cost-effective for limited compute
47	B	RLHF with expert feedback improves safety
48	B	Demonstrates need for domain-specific evaluation
49	A	RL gives whole-output feedback, better for alignment
50	A	All challenges mentioned in the module

Module 5.3: Use of Large Models and Prompt Engineering — 30 Questions

Single Choice Questions (1-10)

Q1 What is the PRIMARY purpose of prompt engineering?

- A. To retrain the entire language model from scratch
- B. To refine prompt phrases using structured text to guide the LLM to produce desired outputs
- C. To reduce the model's parameter count
- D. To replace the need for GPUs

Q2 According to the module, what is the FIRST principle for better using large models?

- A. Use complex technical jargon
- B. Write clear and specific instructions
- C. Keep prompts as vague as possible
- D. Avoid using any examples

Q3 What is the SECOND principle for better using large models?

- A. Give the model time to think
- B. Increase the model size
- C. Use only single-word prompts
- D. Disable the model's reasoning capabilities

Q4 In zero-shot prompting, what does the user provide to the model?

- A. Multiple demonstrations with answers
- B. No demonstrations related to the task result; only the task description
- C. The entire training dataset
- D. A pre-trained model checkpoint

Q5 What is the key difference between zero-shot and few-shot prompting?

- A. Zero-shot uses more parameters
- B. Few-shot provides demonstrations in the prompt to guide the model
- C. Few-shot requires retraining the model
- D. Zero-shot only works for simple math problems

Q6 What is "In-Context Learning" (ICL) as discovered in GPT-3?

- A. A method that requires adjusting all model parameters for each new task
- B. An emerging behavior where the model achieves excellent results with only a few examples without parameter adjustment
- C. A technique for compressing the model
- D. A way to increase the model's training data

Q7 What is the PRIMARY purpose of Chain-of-Thought (CoT) prompting?

- A. To make the model generate shorter answers
- B. To enable complex reasoning by introducing intermediate reasoning steps
- C. To reduce the model's memory usage
- D. To eliminate the need for prompts

Q8 What is the specific prompt added in Zero-Shot-CoT to guide step-by-step thinking?

- A. "Answer immediately"
- B. "Let's think step by step"
- C. "Ignore all previous instructions"
- D. "Use only one word"

Q9 In Few-Shot-CoT, what does each provided example include?

- A. Only the final answer without any explanation
- B. A high-quality inference process showing step-by-step reasoning
- C. Random noise data
- D. Only the question without the answer

Q10 Which of the following is NOT a feature of CoT prompting?

- A. Orderliness: questions broken into ordered steps
- B. Relevance: logical connection between each step
- C. Step-by-step inference: focusing on current step only
- D. Randomness: each step is independent and unrelated

Multiple Choice Questions (11-16)

Q11 Which techniques are recommended for writing clear and specific instructions? (Select all that apply)

- A. Use symbols to clearly distinguish different parts of the prompt
- B. Output structured prompts
- C. Provide a few examples
- D. Use vague and ambiguous language
- E. Keep prompts as short as possible

Q12 Which techniques are recommended to give the model time to think? (Select all that apply)

- A. Develop the steps required to complete the task
- B. Let the model propose a solution first, then compare and assess it
- C. Ask for immediate answers without reasoning
- D. Break complex problems into smaller steps
- E. Disable the model's reasoning capabilities

Q13 What are the benefits of prompt engineering mentioned in the module? (Select all that apply)

- A. Improves LLM capabilities in handling complex tasks like Q&A and arithmetic inference
- B. Helps developers design powerful engineering technologies to connect with LLMs
- C. Enhances LLM security
- D. Enhances LLM capabilities using specialized knowledge and external tools
- E. Reduces the model's parameter count

Q14 Which are advantages of Zero-Shot-CoT over Few-Shot-CoT? (Select all that apply)

- A. Generalization: works across multiple inference tasks with a single template
- B. No manual example design required
- C. Stronger zero-shot inference benchmark for difficult multi-step tasks
- D. Better scalability: performance improves more as model scale increases
- E. Reduced bias from manually designed examples

Q15 Which elements are typically included in few-shot prompting? (Select all that apply)

- A. Task description (optional)
- B. Several examples of task queries and corresponding answers
- C. The final query to be answered by the model
- D. All training data
- E. Model parameter updates

Q16 Which characteristics define Chain-of-Thought prompting? (Select all that apply)

- A. Orderliness: breaking questions into a series of ordered steps
- B. Relevance: close logical connection between each thought step
- C. Step-by-step inference: focusing on current question and related information
- D. Randomness: no connection between steps
- E. Single-step inference: answering everything at once

True or False Questions (17-20)

Q17 Large language models can solve relatively complex problems and have very broad knowledge coverage.

- A. True
- B. False

Q18 In zero-shot prompting, the model is provided with multiple demonstrations related to the task result before being asked to answer.

- A. True
- B. False

Q19 Zero-shot-CoT requires manual design of multi-step inference examples for each task.

- A. True
- B. False

Q20 Few-shot prompts are very sensitive to model scale, while zero-shot-CoT shows better scalability as model size increases.

- A. True
- B. False

Fill in the Blank Questions (21-24)

Q21 To better use large models, two main principles are: (1) _____ clear and _____ instructions, which involves using _____ to distinguish different parts, outputting _____ prompts, and providing a few _____; and (2) Give the model _____ to _____, which involves developing the _____ required to complete the task, letting the model _____ a solution first, then _____ and _____ it.

Q22 Prompt engineering involves _____ prompt phrases using _____ text to guide the _____ to produce _____ outputs. Researchers use it to improve _____ in handling complex tasks such as _____ and _____ inference. Developers use it to design _____ technologies to efficiently connect with _____ or other ecosystem tools. Users can improve _____ security through prompt engineering and enhance capabilities by using _____ knowledge and external _____.

Q23 Zero-shot prompting is a technique where the user does not provide _____ related to the task result, and directly prompts the language model to provide _____ answers. However, on more _____ tasks, zero-shot prompting performs poorly. Few-shot prompting provides _____ in the prompt to guide the model. GPT-3 researchers discovered _____ (ICL), which does not require model _____ adjustment and can achieve excellent results with only a few _____ of downstream tasks.

Q24 Chain-of-Thought (CoT) prompting enables _____ reasoning by introducing _____ reasoning steps. It has two main types: _____-shot-CoT and _____-shot-CoT. CoT features include _____ (breaking questions into ordered steps), _____ (logical connection between steps), and _____ inference. Zero-shot-CoT adds the prompt " _____ " to guide the model. Few-shot-CoT provides examples with _____-quality _____ processes.

Drag and Drop / Matching Questions (25-30)

Q25 Match each principle to its corresponding technique:

Principle	Technique
-----------	-----------

- | | |
|-----------------------------|--|
| A. Write clear instructions | 1. Develop steps, let model propose solution first |
| B. Give model time to think | 2. Use symbols, structured prompts, provide examples |

Write your answer as: A→?, B→?

Q26 Match each prompt type to its description:

Prompt Type	Description
-------------	-------------

- | | |
|------------------|--|
| A. Zero-shot | 1. Provides demonstrations to guide the model |
| B. Few-shot | 2. No demonstrations, direct task prompting |
| C. Zero-shot-CoT | 3. Adds "Let's think step by step" |
| D. Few-shot-CoT | 4. Provides examples with step-by-step reasoning |

Write your answer as: A→?, B→?, C→?, D→?

Q27 Match each CoT feature to its description:

Feature	Description
---------	-------------

- | | |
|---------------------------|---|
| A. Orderliness | 1. Focus on current question, approach final answer gradually |
| B. Relevance | 2. Questions broken into a series of ordered steps |
| C. Step-by-step inference | 3. Close logical connection between each thought step |

Write your answer as: A→?, B→?, C→?

Q28 Match each advantage to its description:

Advantage	Description
-----------	-------------

- | | |
|-----------------------------|---|
| A. Generalization | 1. Works across multiple tasks with single template |
| B. No manual example design | 2. Reduces prompt engineering workload |
| C. Better scalability | 3. Performance improves more with larger models |
| D. Reduced bias | 4. Less bias from manually designed examples |

Write your answer as: A→?, B→?, C→?, D→?

Q29 Match each method to its best use case:

Method	Use Case
A. Zero-shot	1. Complex multi-step reasoning with examples
B. Few-shot	2. Simple tasks where model has existing knowledge
C. Zero-shot-CoT	3. Tasks needing demonstrations for better performance
D. Few-shot-CoT	4. Complex reasoning without designing examples

Write your answer as: A→?, B→?, C→?, D→?

Q30 Match each In-Context Learning property to its description:

Property	Description
A. No parameter adjustment	1. Model predicts next word based on current input
B. Few examples needed	2. Only task description and examples provided
C. Autoregressive nature	3. Achieves results without retraining
D. Task packaging	4. Minimal examples for downstream tasks

Write your answer as: A→?, B→?, C→?, D→?

Scenario-Based Questions (31-35)

Q31 A user wants to classify customer reviews as Positive or Negative. The model has been trained on general text and understands sentiment.

Which prompting approach is MOST appropriate?

- A. Few-shot-CoT with 5 step-by-step reasoning examples
- B. Zero-shot prompting with clear instruction: "Classify this review as Positive or Negative"
- C. Zero-shot-CoT with "Let's think step by step"
- D. Provide the entire training dataset in the prompt

Q32 A student needs help solving: "A train travels 120 km in 2 hours, then 80 km in 1 hour. What is the average speed?" The model sometimes gives wrong answers for multi-step problems.

Which approach is MOST likely to produce the correct answer?

- A. Zero-shot: "What is the answer?"
- B. Zero-shot-CoT: "Let's think step by step. Calculate total distance and total time, then divide."
- C. Few-shot with only final answers: "Q: 2+2=?, A: 4. Q: [problem], A:"
- D. Ask the model to guess randomly

Q33 A company is building a customer service chatbot. They want to prevent the model from generating harmful or biased content.

Which prompt engineering technique should they prioritize?

- A. Use vague prompts to confuse the model
- B. Design structured prompts with clear safety guidelines and use prompt engineering to improve LLM security
- C. Disable all safety features
- D. Train the model from scratch without any safety data

Q34 A developer needs to solve three different tasks: sentiment analysis, math word problems, and code debugging. They want to use the same prompting approach for all three without designing specific examples for each.

Which method is MOST suitable?

- A. Few-shot-CoT with custom examples for each task
- B. Zero-shot-CoT with "Let's think step by step" template
- C. Full model retraining for each task
- D. Random prompting

Q35 A team is testing prompting methods on two models: a 7B parameter model and a 70B parameter model. They observe that one method shows much better improvement on the larger model.

Which method is likely showing this scalability pattern?

- A. Few-shot prompting
- B. Zero-shot-CoT
- C. Random guessing
- D. No prompting at all

Advanced Concept Questions (36-40)

Q36 GPT series models are autoregressive language models. What does this mean in terms of text generation?

- A. They generate all tokens simultaneously in parallel
- B. They predict the next word based on the current input, then combine the prediction with input to predict the next word, repeating cyclically
- C. They only generate the first word and stop
- D. They require all future tokens to be known in advance

Q37 What is the fundamental difference between In-Context Learning (ICL) and traditional fine-tuning?

- A. ICL requires adjusting all model parameters; fine-tuning does not
- B. ICL does not require model parameter adjustment and works with only a few examples; fine-tuning updates model parameters
- C. ICL only works for small models; fine-tuning works for large models
- D. There is no difference between them

Q38 Why does Zero-Shot-CoT outperform standard zero-shot prompting on complex multi-step tasks?

- A. Because it adds more parameters to the model
- B. Because the "Let's think step by step" prompt triggers the model to generate intermediate reasoning steps, improving accuracy
- C. Because it uses a different model architecture
- D. Because it requires less computation

Q39 Why is prompt engineering often preferred over retraining for adapting LLMs to new tasks?

- A. Because retraining is always impossible
- B. Because prompt engineering is faster, cheaper, and does not require modifying model parameters while achieving good results
- C. Because prompt engineering increases model size
- D. Because retraining is always more accurate

Q40 Which statement best describes the evolution of prompting techniques from simple to advanced?

- A. Zero-shot → Few-shot → Chain-of-Thoughts (Few-shot-CoT and Zero-shot-CoT), with each step adding more reasoning structure and guidance
- B. Chain-of-Thoughts → Zero-shot → Few-shot
- C. All prompting techniques are identical
- D. Advanced prompting makes simple prompting obsolete

Q	Answer	Explanation
1	B	Prompt engineering refines prompts to guide LLM outputs
2	B	First principle: write clear and specific instructions
3	A	Second principle: give the model time to think
4	B	Zero-shot: no demonstrations, only task description
5	B	Few-shot provides demonstrations; zero-shot does not
6	B	ICL: excellent results with few examples, no parameter adjustment
7	B	CoT enables complex reasoning via intermediate steps
8	B	Zero-shot-CoT uses "Let's think step by step"
9	B	Few-shot-CoT includes high-quality inference process
10	D	Randomness is NOT a CoT feature
11	A, B, C	D, E are incorrect; clarity and structure help
12	A, B, D	C, E are incorrect; reasoning needs time and steps
13	A, B, C, D	E is false; prompt engineering doesn't reduce parameters
14	A, B, C, D, E	All are advantages of Zero-shot-CoT
15	A, B, C	D, E are incorrect; few-shot uses examples, not full data
16	A, B, C	D, E are incorrect; CoT requires order and relevance
17	A - True	Large models have broad knowledge and complex problem-solving
18	B - False	Zero-shot provides NO demonstrations
19	B - False	Zero-shot-CoT does NOT require manual examples
20	B - False	Few-shot is NOT very sensitive; zero-shot-CoT scales better
21	Write, specific, symbols, structured, examples, time, think, steps, propose, compare, assess	Two principles for better LLM use
22	refining, structured, LLM, desired, LLM capabilities, Q&A, arithmetic, engineering, LLMs, LLM, specialized, tools	Prompt engineering definition and benefits
23	demonstrations, task-related, complex, demonstrations, In-Context, Learning, parameter, examples	Zero-shot, few-shot, and ICL
24	complex, intermediate, Few, Zero, Orderliness, Relevance, step, step, Let's think step by step, high, inference	CoT types and features
25	A→2, B→1	Clear instructions = symbols/structure/examples; Time to think = steps/propose/compare
26	A→2, B→1, C→3, D→4	Zero-shot=no demos, Few-shot=demos, Zero-shot-CoT=step by step, Few-shot-CoT=reasoning examples
27	A→2, B→3, C→1	Orderliness=ordered steps, Relevance=logical connection, Step-by-step=gradual approach
28	A→1, B→2, C→3, D→4	Generalization=single template, No design=less workload, Scalability=model size, Reduced bias=less example bias
29	A→2, B→3, C→4, D→1	Zero-shot=simple tasks, Few-shot=demonstrations needed, Zero-shot-CoT=complex without examples, Few-shot-CoT=complex with examples
30	A→3, B→4, C→1, D→2	No param adjustment=without retraining, Few examples=minimal data, Autoregressive=next word prediction, Task packaging=description + examples
31	B	Simple classification: zero-shot with clear instruction
32	B	Complex math: zero-shot-CoT triggers step-by-step reasoning

33	B	Security: structured prompts with safety guidelines
34	B	Cross-task: zero-shot-CoT generalizes without custom examples
35	B	Zero-shot-CoT shows better scalability with model size
36	B	Autoregressive: predict next word cyclically
37	B	ICL: no parameter adjustment; fine-tuning: updates parameters
38	B	"Let's think step by step" triggers intermediate reasoning
39	B	Prompt engineering is faster, cheaper, no parameter changes
40	A	Evolution: zero-shot → few-shot → CoT variants